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NSP/004/120 (OHI 20) Guidance on mounting overhead line plant and equipment on HV poles

1. Purpose

The purpose of this document is to provide guidance on the approved arrangement drawings detailing the mounting of additional equipment onto HV structures used on the CE Electric UK distribution system.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
OHI 20	Feb 1982	OHI 20 Erection of plant & equipment operating at voltages of 11, 20 & 33kV mounted on wood poles.
0.309 – Appendix 3	March 1996	High Voltage Single Circuit Overhead Lines On Wood Poles For Voltages Up To And Including 33 kV

2. Scope

This document covers all plant and equipment required to be mounted onto HV structures. It provides a range of detailed approved drawing arrangements indicating the agreed mounting position for the plant and equipment. Additionally each arrangement includes an associated pole fabrication drawing for the wood pole fabricator.

3. Technical Specification

3.1 Background

An HV overhead line may require on or more of the following items of plant/equipment to be mounted onto a support:-

- a) Pole Mounted Aerial Switches – ABSD
- b) Drop out expulsion fuses or intelligent sectionalisers -
- c) Pole Mounted Power transformers - PMT
- d) Pole Mounted Automatic Reclosers – PMAR
- e) Surge Diverters
- f) Cable terminations
- g) Triggered Spark Gaps
- h) Fault passage Indicators
- i) 400A LV Fuse Cut-outs

To ensure we maintain a standard design approach across CE Electric UK, a range of detailed assembly drawings catering for standard arrangements have been prepared. Each drawing has been developed to take into account the ESQCR regulations, Operational requirements, & any possible Live Line Working requirements.

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Appendix 1 tabulates the approved arrangement drawings that must be utilised and the drawing reference numbers that must be included on any line records or schedules. No other pole arrangements shall be erected without prior agreement from the overhead line standards manager.

3.1.1 Procedure for introducing new pole arrangements

Where pole arrangements other than those detailed in Appendix 1 are required. Copies of proposed arrangement drawings must be submitted to the overhead line standards manager for approval. Where the proposed arrangement is acceptable the drawing will be given an agreed drawing number which must be referenced on any future line schedule.

3.2 Pole Mounted Aerial Switches – ABSD

ABSD's shall be designed and supplied in accordance with the company technical specification NPS/001/003.

The current design of ABSD's have been developed to allow them to be mounted onto unearthed poles thus improving the lightning performance of the structure, and removing the need to install/inspect the previously associated earth mat arrangements.

Key Features

- ABSD's are now designed to be operated using insulated fuse rod operating sticks.
- With respect to ground level the operating mechanism has been fixed on all pole heights @ 4.6m from ground level ensuring that all switches can be operated using 3 insulated rods.
- Poles up to and including 14.0m in height can be accommodated without a requirement to utilise pilot pins.
- ABSD can only be mounted in the horizontal orientation.
- ABSD's incorporate springs to make the units fully independent manual with 10kA fault make capabilities.
- All jumpers and connections shall be in accordance with OHI 7.
- Switches incorporate a safety lock arrangement to allow the units to be used as points of isolation.
- See drawing 1091471525 for switch details.

[Manufactures Mounting Instructions](#)

The attached link provides additional information on the ABSD.

3.3 Drop out expulsion fuses or intelligent sectionalisers -

Drop out expulsion fuses and intelligent sectionalisers shall be designed and supplied in accordance with the company technical specification NPS/001/004.

Under normal circumstances fuse isolators shall be arranged such that, after operation of a fuse, the jumpers that remain live will normally be those that are attached to the top of the fuse carrier. Where it is considered advantageous the fuse isolators may be mounted one pole or more out from the tee-off support, whichever gives the most convenient operating conditions.

Fuse isolators shall be mounted a minimum of 5.1m from ground level and a maximum of 6.5m (equivalent to a maximum of 5 insulated rods) from ground level. All associated jumpers must maintain a minimum clearance of 4.6m from the ground increasing to 5.8m where the equipment is located adjacent to roads.

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Where pilot pins are mounted to provide jumper support at 20kV, the exposed metalwork of the insulator pins shall be protected through the use of tubular bird guards.

3.3.1 Drop Out Fuse isolators

Fey Features

- A polymeric back portion suitable for use at 11 or 20kV see drawing 1091471503 sheet 1 for details.
- Units can utilise fuse carriers, solid links or re-settable intelligent sectionalisers.
- Units incorporate a bypass capability to allow a temporary shorting device to be installed while fuse links/sectionalisers are interchanged by live line techniques.

3.3.2 Fuse Elements

See drawing 1091471503 sheet 3 or 3a for details of available fuse elements

3.3.3 Intelligent sectionalisers

Auto-sectionalisers discriminate between a transient & a persistent fault by counting the passage of fault current during the auto-reclose sequence. They operate during the dead time of the auto-reclose sequence, after a pre-determined number of passages of fault current. They shall only be fitted beyond multi-shot auto reclose circuit breakers. The rational behind this policy is to preserve the integrity of the main lines under fault conditions.

See drawing 1091471504 sheet 1 for details on the intelligent sectionalisers.

- Sectionalisers are suitable for installation on multi voltages ie 6kv, 11kv or 20kv.
- It is intended that the sectionalisers will always be fitted to Linegear 2000 back portions regardless of the condition of the existing J and P back portion.
- Sectionalisers are available with an operating sequence of 1,2 or 3 "counts" of fault current.
- It is permissible to use 3 shot sectionalisers in series with 2 shot sectionalisers, in order to get improved sub spur protection, however it must be remembered in this case that the multi shot reclosing circuit breaker must be set for four trips to lock out. The sectionaliser should be set to operate in at least one less count than the backup recloser. Where sectionalisers are used in series, the downstream sectionaliser should have one less count than the upstream sectionaliser. It is not intended that there will be more than two sectionalisers in series.
- The sectionalisers will be colour coded for current rating and number of shots, this will make identification possible from ground level.

3.4 Pole Mounted Power transformers – PMT

Pole mounted transformers shall be designed and supplied in accordance with the company technical specification NPS/003/011.

All transformers

To help protect the transformers against transient overvoltages on the HV side, all transformers are fitted with duplex spark gaps. The gaps shall be set as follows:-

2 x 25mm @ 11kV
2 x 38mm @ 20kV

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3.4.1 Jumpers and connections to pole mounted transformers

LV Connections

Connections between the LV terminals of pole mounted transformers and LV fuse cut-outs shall be provided through the use of PVC insulated and sheathed, single core, annealed copper conductors. Connections to the LV bushings shall be via single hole fixing compression lugs with holes appropriate to the stem diameter through the bushing

Transformer Rating	Three phase current requirement in Amps to match TX KVA @433V	Reqd Cable Size	Standard LV Bushing Size	Dia. hole reqd in lug connection
50kVa or below	67A	35mm2	Type A (250A) M12	14mm
100 kVA	133 A	95mm2		
200 kVA	267A		120mm2	Type B (630A) M20
315 kVA	420A			

HV Connections

Shall be provided through the use of 32mm² (7/2.46mm) Hard drawn (black) PVC insulated jumpers terminated onto the M12 HV studs. See OHI 7 for further information on connections to overhead lines

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3.4.2 Key dimensional data associated with pole mounted transformers

Description	Total Mass of Transformer (kg)	Dimensions Length	Width	Height	Single Bolt Fix or Platform	Min - Single Pole or 'H' pole when installed in main line	Min - Single Pole or 'H' pole when installed on totem or inverted pole
11kV Pole Mounted Transformers							
25kva-11/250v-1Phase - Pole Mounted Transformers	220	650	630	1145	SB	S	S
50kva-11/250v-1Phase - Pole Mounted Transformers	365	715	695	1005	SB	S	S
100kva-11/250v- 1Phase - Pole Mounted Transformers	535	795	725	1135	P	S	S
25kva-11/433v-3Phase - Pole Mounted Transformers	320	800	490	1010	P	S	S
50kva-11/433v-3Phase - Pole Mounted Transformers	330	935	485	1025	P	S	S
100kva-11/433v-3Phase - Pole Mounted Transformers	475	1045	645	1155	P	S	S
200kva-11/433v-3Phase - Pole Mounted Transformers	795	1215	785	1185	P	S	H
315kva-11/433v-3Phase - Pole Mounted Transformers	1035	1265	965	1150	P	H	H
20kV Pole Mounted Transformers							
16kva-20/250V-1Phase - Pole Mounted Transformers	185	650	630	995	SB	S	S
25kva-20/250V-1Phase - Pole Mounted Transformers	245	650	630	1085	SB	S	S
50kva-20/250V-1Phase - Pole Mounted Transformers	365	685	705	1025	SB	S	S
100kva-20/250v-1Phase - Pole Mounted Transformers	545	835	725	1160	P	S	S
50kva-20/433V-3Phase - Pole Mounted Transformers	510	965	605	1230	P	H	S
100kva-20/433v-3phase - Pole Mounted Transformers	680	1085	565	1245	P	H	H
200kva-20/433v-3phase - Pole Mounted Transformers	945	1205	645	1305	P	H	H
315kva-20/433v-3phase - Pole Mounted Transformers – under consideration	Tba	Tba	Tba	Tba	P	H	H

Notes relating to the logic used in the formulation of the above tables.

- 1) ENA TS 35-1 allows for single bolt transformers up to 50kva provided weight does not exceed 400kg
- 2) ENA TS 43-95 allows single pole platform mounted transformers up to 100kVa or 1000kg. Due to the general reduction in weight of pole mounted transformers we are using the 1000kg criteria as the major design driver provided the overall envelope size of the transformer has not increased significantly.
- 3) ENA TS 43-95 allows for 'H' pole platform mounted transformers up to 200kVa or 1400kg. The additional special YEDL steelwork rated at 1750kg will no longer be required for new transformers.
- 4) To maintain the RYB left to right phasing on 20kV terminals (back to the line source) this requires all terminal arrangements to be mounted on 'H' poles.
- 5) A risk assessment approach has been applied to define different criteria for transformers located on intermediate or stayed supports compared with those located on unaided self supporting supports. This results in the table reflecting a greater use of 'H' pole platform mounted structures (where the weight is distributed about the centreline of the support compared with a cantilever load on single pole platforms). This policy aims to try to reduce the incidence of leaning poles on totem or inverted s/s supports. The application of stays to the pole side of inverted poles shall cease as a means of trying to keep poles straight as they rarely correct the situation and may create additional electrical hazards.

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3.5 Pole Mounted Automatic Reclosers – PMAR

Pole Mounted Automatic Reclosers shall be designed and supplied in accordance with the company technical specification NPS/001/009.

All new PMAR's installations shall installed in accordance with drawing 1091091310 sheet 10 with all jumpers and connections arranged as detailed in NSP/004/107.

All new PMAR installations shall be provided with an "In-Line Isolator" as shown on drawing 1091010752 located in the HV conductors on the load side of the installation. This unit will enable the recloser site to be used as a point of isolation without the need to cut or temporarily remove jumpers.

New PMAR installations must incorporate fully rated live line section insulators. They must not be reliant upon the tension insulators across section poles.

Approved jumper arrangements shall be as detailed in Appendix 1 & 2.

Modern PMAR's utilise low level control cabinets that contain remote control equipment and battery chargers.

The two 12 Volt, 7.2Ah rechargeable batteries installed in the control cabinet must be charge cycled several times before they are installed into new control cabinets. This process is carried out by the suppliers before they are dispatched, however long transportation periods and storage delays can result in the batteries going into a discharged state. Lead acid batteries will not perform as planned unless they are correctly conditioned.

The intended mode of operation for this equipment in the event of a loss of auxiliary supplies has been designed to be as follows:

- i) Full powered operation of the relay and radio communication for 4 hours
- ii) After this time the internal timer will shut down the radio reducing the power drain from 320mA to 70mA (failure to shut down the radio will result in full battery discharge in 22 hours).
- iii) Theoretically the Nulec CAPM controller should, therefore, be able to be powered up for another 80 hours (3.3days). During this remaining period full protection and recloser operation will be available.

The above scenario is only available if the batteries are in good condition when the units are commissioned. The CAPM controller has no deep discharge protection to protect the batteries, hence if the unit is left switched on without the external power source the batteries will eventually fall into their deep discharged state. This has an adverse effect on the capacity of the battery if they are left in this state for a long time. The battery can be cycled in order to reverse the process if this has happened but this will typically take 3 cycles spread over several days.

Where it is known that auxiliary supplies will not be available to a commissioned unit for long periods of time, arrangements shall be made to isolate the LV supplies in the control cabinet to protect the battery from going into deep discharge.

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3.6 Surge Diverters

Surge Diverters shall be designed and supplied in accordance with the company technical specification NPS/001/008.

To ensure effective operation of surge diverters they shall be installed as close as possible to the cable termination. (Modern diverters act as a substitute for stand-off / support insulators used in cable terminations). HV and earth connections shall be as short and straight as possible, with the diverter earth terminal being bonded directly to the HV cable earth terminal then taken down to an earth electrode. The combined resistance to earth of the electrode and cable sheath for surge diverter installations shall not exceed 10ohms.

- Surge diverters shall normally be installed in all cable terminations.
- Surge diverters must be disconnected before cables are overvoltage tested.

Description	Drawings Number
11kV surge diverter	1091480162 sht 2
20kV surge diverter	1091480162 sht 3
33kV surge diverter	1091480165 sht 3 or sht 4
66kV surge diverter	1091480166 sht 1& sht 2

Further guidance on the company's application policies can be found in IMP/007/011 "Code of practise for the application of Lightning protection"

3.7 Cable terminations

Cable terminations shall be designed and supplied in accordance with the company technical specification NPS/002/015 & NPS/002/014.

HV cables of polymeric triplex formation shall be terminated onto overhead supports using a cable termination support bracket as detailed on 1181130009 sheet 6.

Cables shall be secured to supports using clips or cleats as appropriate. A spacing of approximately 450 - 600mm shall be adopted for LV & HV cables respectively.

All cables shall have a cable guard fitted to protect the cable on the pole and reduce the risk of it being utilised as a climbing aid. The guard shall extend for a distance of 3.15m from the ground and protrude into the ground to a depth of 150mm. See drawing no. 1091010171 for details.

Note – Where a cable is controlled by expulsion fuses, it is company policy to arrange the expulsion fuses so that when they operate, the fuse hanging in the mounting is dead.

3.7.1 Earthing conductors

Earthing & bonding conductors shall normally be provided through the use of 32mm² (7/2.46mm) Green PVC insulated copper conductor. It shall normally be terminated onto the steelwork or plant using the appropriate compression lug.

(Note PMAR's, ABSD's and all earthed poles operating at EHV shall utilise 70mm² conductor.

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The earthing conductors shall be cleated or stapled to the pole at nominal intervals of 500mm between fixings. All earth leads shall be protected using cable guards See drawing no 1091010149 sht3 for details

3.8 Triggered Spark Gaps

Triggered Spark Gaps shall be designed and supplied in accordance with the company technical specification NPS/001/006.

Triggered spark gaps shall be mounted in accordance with drawing 1091200014.

To eliminate transient earth faults and retain supplies to customers during permanent earth faults, a tuned arc suppression (Petersen) coil reactor earthing method may be adopted on selected rural HV systems in areas subject to high lightning activity.

Where Arc Suppression coil earthing is applied, and unless specified otherwise, yellow phase triggered spark gaps should be installed to avoid phase to phase fault auto reclose operations during lightning activity. The frequency of installation for triggered spark gaps shall be in accordance with the company's code of practice IMP/007/011.

3.9 Fault passage Indicators – FPI's

Fault passage indicators shall be designed and supplied in accordance with the company technical specification NPS/001/014.

Modern FPI's contain the ability to identify the fault status locally via a visual flashing light and simultaneously send a signal to a centralised remote monitoring centre via an inbuilt GSM communications system.

3.9.1 Operating Modes

FPI's are suitable for recording permanent, intermittent and transient faults.

A passage of fault current will result in a red 'super' bright LED or Xenon light to flash for a designated period of time (as detailed in section 3.9.2 and 3.9.3) depending upon the fault type. This time period will provide sufficient time for field staff to inspect the indicators on the system and patrol the faulty section of network. SEF faults are not indicated unless loss of voltage is also detected. This is to prevent spurious operation due to load changes.

3.9.2 Permanent Fault

If the fault is permanent then any upstream FPI's will alarm and flash the Xenon light until the voltage is restored or for a period of 3 hours, whichever occurs first. To save battery capacity, once the Xenon light stops flashing the red LED or low energy identifier will commence flashing and continue for 24 hours. At the time of the fault, the GSM module will signal back to the basestation to record the permanent fault event. An additional signal will also be generated when the units are reset and normal system voltage is restored.

3.9.3 Intermittent and Transient Faults

In the instances where there has been a successful auto-reclose and the system volts have been restored, only the red LED will flash for a period of 24 hours. The Xenon lamp should have been reset by the system voltage after 60 seconds. At the time of the fault event, the GSM module should signal back to the basestation to record the transient fault event.

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3.9.4 Installation

FPI's shall normally be installed on clean intermediate or angle section poles approx 3.5m below the conductors. Ideally the selected supports will be adjacent to aerial switches, or solid spurs. FPI's work on the principle of sampling the magnetic field given out from the conductors, hence to ensure reliable operation they must be kept away from other sources of magnetic fields.

Local knowledge of road crossings etc should be used for the final choice of location.

Location in the network where FPI's should not be installed on supports

- a) With underground cables
- b) With transformers
- c) With double circuit lines
- d) With tee off lines
- e) Closer than 300m to adjacent 275-400kV lines
- f) Closer than 150m to adjacent 132kV lines
- g) Closer than 100m to adjacent 66kV lines
- h) Closer than 50m to adjacent 33kV lines
- i) Closer than 30m to adjacent 20/11kV lines
- j) Closer than 100m to rail track HV overhead conductors

3.10 400A LV Fuse Cut-outs

400A LV Fuse Cut-outs shall be designed and supplied in accordance with the company technical specification NPS/002/006.

LV fuse cut-outs shall be fixed to wood pole supports using M12 & 85mm coach screws at a minimum height from ground level of 3660mm with a nominal 305mm spacing between units. The units must be mounted to ensure there is no hazard to the operator from the ACD (Anti Climbing Devices) during fuse removal. The cut-outs are supplied complete with M12 X 35mm bolts to allow the LV tails to be terminated using an appropriate sized copper compression lug.

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4. References

4.1 External Documentation

Reference	Title
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4.2 Internal documentation

Reference	Title
IMP/007/011	Code of practise for the application of Lightning protection
NPS/004/042	HV Single circuit overhead lines on wood poles up to and including 33kV
NPS/001/003.	Technical Specification for Pole Mounted Air Break Switch Disconnectors.
NPS/001/004	Technical Specification for 11, 20 & 33kV pole mounted expulsion fuse's, automatic sectionalising links and solid links
NPS/001/006	Technical Specification for Overhead Line insulators
NPS/001/008	Technical Specification for surge arrestors
NPS/001/009	Technical Specification for 11, 20 & 33kV Pole Mounted Auto Reclose Circuit Breakers
NPS/001/014	Technical Specification for Overhead Line fault passage indicators
NPS/002/006	Technical Specification for Heavy Duty and House service cut-outs
NPS/002/014	Technical Specification for LV Joints and terminations
NPS/002/015	Technical Specification for 11 & 20kV Joints and terminations
NPS/003/011	Technical Specification for 11 & 20kV Distribution Transformers

5. Definitions

Term	Definition
ABSD	Air Break Switch Disconnectors

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	04/01/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	08/01/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	08/01/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	03/01/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
S Gardner	OHL Construction Manager	S Gardner	23/03/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Neil Denley	Primary Engineering Projects Manager	Neil Denley	27/03/07

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Appendix 1

Approved Pole Mounted Plant Arrangement Drawings

	Drawing No.	
Arrangement	11 kV	20 kV
Std Cable Termination Poles		
Cable Laypole	1.00.043.4009 sht 5	
Cable Terminal (single)	1.00.043.4010 sht 6	
Cable Terminal (H Pole)	1.00.043.4011 sht 6	
Std S/S Poles		
Single Phase		
Single Pole Term. Trans, Platform or Single bolt fixing, without fuses	1.09.107.0810 sht 1	
Single Laypole Trans, Platform or Single bolt fixing , without fuses	1.09.107.0811 sht 1	
Single Straight Line Pole, Trans, Platform or Single bolt fixing, without fuses	1.09.107.0814 sht 1	
Single Pole Term, In Line or Laypole Trans, with or without fuses	1.09.108.0821 sht 3	Not Available
Single Pole Totem, Trans, with HV Fuses	1.09.107.0819 sht 3	
Three Phase		
Single Pole Term.Trans, Platform Mounted, without fuses	1.09.107.0810 sht 2	Not Available
Single Laypole Trans, Platform mounted without fuses	1.09.107.0811 sht 2	
Single Straight Line Pole, Trans, Platform mounted without fuses	1.09.107.0814 sht 2	
11 kV Terminal S/S, Line or Laypole S/S - Three Phase with or without fuses	1.09.108.0821 sht 1	Not Available
'H' Pole Terminal, Trans, Platform Mounting without fuses	1.09.108.0860 sht 1	
'H' Laypole, Trans, Platform Mounting without fuses	1.09.108.0860 sht 3	
'H' Terminal Pole, Trans, Platform Mounting c/w fuses	1.09.108.0860 sht 5	
'H' Totem Pole, Trans, Platform Mounting c/w fuses	1.09.108.0862 sht 3	
Short Leg 'H' Pole Terminal or in line, Trans, complete with Fuses	1.09.108.0864 sht 1	
Short Leg 'H' Pole Term or in line, Trans, Short leg 90° to line c/w fuses.	1.09.108.0864 sht 2	
Single Totem Pole, Trans, Platform Mounted	1.09.108.0865 sht 1	
ABSD Poles	11 kV	20 kV
ABSD mounted on Single Intermediate pole	1.09.109.0906 sht 7	
ABSD mounted on 'H' Intermediate pole	1.09.109.0908 sht 1	
ABSD mounted on Lay Terminal 'H' Pole	1.09.109.0906 sht 3	
ABSD mounted on Lay Terminal Single Pole	1.09.110.0933 sht 1	
ABSD mounted on Single Intermediate pole with Fuse Isolators and Cable Termination	1091100932 sht 5	
ABSD mounted on Single Terminal Pole with Cable Termination	1.09.110.0937 sht 5	
ABSD mounted on H Terminal Pole with Cable Termination	1.09.109.0907 sht 11	
Auto Recloser Poles		
PMAR (Nulec) on Single Pole with In-Line Isolators (including Cast Resin VT's)	1.09.109.1310 sht 10	
PMAR (Nulec) on Single Pole (including Oil Filled VT)	1.09.109.1310 sht 7	

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Fuse Isolator Poles	11 kV	20 kV
Three Phase Tee Off to Laypole with HV Fuses (three fuses one side of pole)	1.09.112.1063 sht 1	
Three Phase Tee Off to Laypole with HV Fuses (two fuses one side of pole)	1.09.112.1061 sht 1	
Three Phase Full Span Tee off with HV Fuses	1.09.112.1064 sht 1	
Three Phase Straight Line Section pole with HV Fuses	1.09.112.1067 sht 1	
Single Phase Solid Tee Off with HV Fuses	1.09.111.1003 sht 1	
Three Phase Cable Term Pole Controlled by HV Fuses (Supply O/H to U/G)	1.00.043.4010 sht 10	
Three Phase Cable Term Pole Controlled by HV Fuses (Supply U/G to O/H)	1.09.112.1069 sht 1	
Three Phase H Cable Term Pole Controlled by HV Fuses (Supply O/H to U/G)	1.09.112.1069 sht 15	
Three Phase Intermediate Pole with cable term Controlled by HV Fuses (Supply O/H to U/G)	1.09.112.1069 sht 7	
Single Phase Tee Off to Laypole with HV Fuses	1.09.111.1000 sht 1	